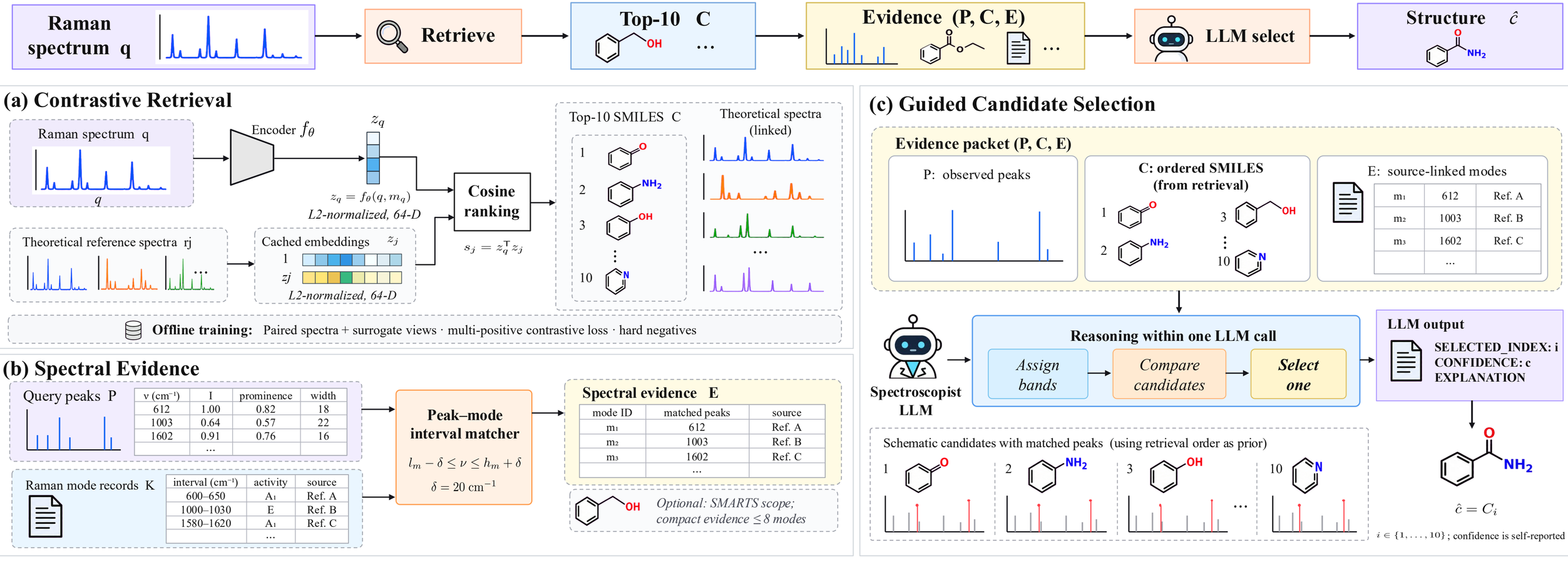

Raman spectrum q
Retrieve
Top-10 C
Evidence (P, C, E)
LLM select
Structure ĉ
(a) Contrastive Retrieval
Raman spectrum q
q
Encoder f_θ
z_q
$z_q = f_\theta(q, m_q)$
L2-normalized, 64-D
Theoretical reference spectra rj
Cached embeddings z_j
1
zj
L2-normalized, 64-D
Cosine ranking
$s_j = z_q^\top z_j$
Top-10 SMILES C
Theoretical spectra (linked)
Offline training: Paired spectra + surrogate views · multi-positive contrastive loss · hard negatives
(b) Spectral Evidence
Query peaks P
ν (cm⁻¹) | I | prominence | width
612 | 1.00 | 0.82 | 18
1003 | 0.64 | 0.57 | 22
1602 | 0.91 | 0.76 | 16
Raman mode records K
interval (cm⁻¹) | activity | source
600–650 | A₁ | Ref. A
1000–1030 | E | Ref. B
1580–1620 | A₁ | Ref. C
Peak–mode interval matcher
$l_m - \delta \le \nu \le h_m + \delta$
$\delta = 20\ \mathrm{cm}^{-1}$
Spectral evidence E
mode ID | matched peaks | source
m₁ | 612 | Ref. A
m₂ | 1003 | Ref. B
m₃ | 1602 | Ref. C
Optional: SMARTS scope; compact evidence ≤ 8 modes
(c) Guided Candidate Selection
Evidence packet (P, C, E)
P: observed peaks
C: ordered SMILES (from retrieval)
E: source-linked modes
m₁ | 612 | Ref. A
m₂ | 1003 | Ref. B
m₃ | 1602 | Ref. C
Spectroscopist LLM
Reasoning within one LLM call
Assign bands
Compare candidates
Select one
LLM output
SELECTED_INDEX: i
CONFIDENCE: c
EXPLANATION
Schematic candidates with matched peaks (using retrieval order as prior)
$\hat{c} = C_i$
$i \in \{1, \ldots, 10\}$; confidence is self-reported